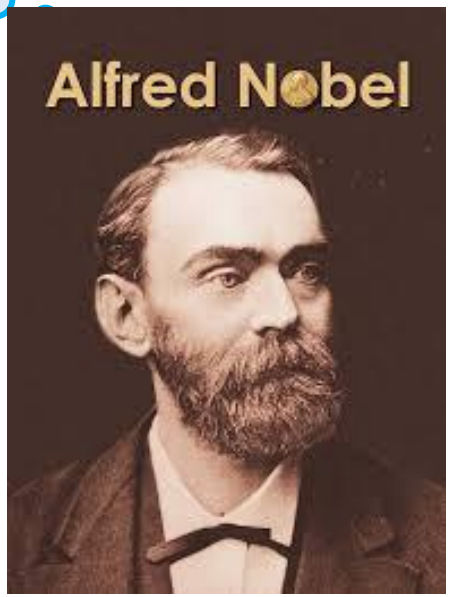
CY4001: Industrial Chemistry

Unit 2: Industrial Safety & Hazards Management

Alfred Nobel

Industrial Chemist



The man behind the Nobel prize

Businessman, chemist, engineer, inventor, and a philanthropist

- Alfred Nobel obtains the first patent on nitroglycerin (blasting oil) as an industrial explosive. He later made dynamite.
- A French newspaper mistakenly published an obituary for him while he was still alive, calling him a ***"merchant of death"***
- He wanted to be remembered for his contributions to science and peace
- Nobel believed that people could improve society through science, knowledge, and humanism.

Unit 2: Industrial Safety & Hazards Management

Introduction

Industrial safety management is a multi-disciplinary approach to creating and maintaining a safe and efficient work environment. It involves reducing, controlling, and eliminating hazards from industrial units.

Safety topics

- General Safety
- Occupational Safety and Health
- Process and Production Safety
- Material Safety
- Workplace Safety
- Fire Safety
- Electrical Safety
- Building and Structural Safety

Causes of accidents

- Process deviations
- Parameters with regard to the state of the substance
- Proximity to other toxic substances.
- Runaway reaction
- Hardware failure
- Boiling Liquid
- Expanding Vapor
- Explosion on the chemicals during transportation.

Major Consequences

- Loss of life / injuries
- Impact on livestock
- Damage to Flora/fauna
- Environmental Impact (air, soil, water)
- Financial losses to the industry

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Industrial Hygiene

Industrial hygiene is the science and art of protecting and enhancing the health and well-being of workers. The field of industrial hygiene tackles a wide range of job-related hazards including environmental, chemical, physical, and ergonomic hazards.

5 Principles of Industrial Hygiene

1.Anticipation: a survey of the workplace design, operations, processes, work tasks, materials and worker population, a current inventory of chemicals and their Safety Data Sheets

2.Recognition: Recognizing and understanding the potential hazards of the work environment, internal processes, and job tasks

3.Evaluation: Collection of data through various methods such as air sampling, personal monitoring, and biological monitoring.

4.Control Over Worker Exposure: implementing measures to minimize or eliminate workers' exposure to hazardous substances in the workplace

5.Confirmation of Control Measures: Assessing the performance of hazard control measures and subsequent worker exposures

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Hazard Identification and Risk Assessment

- Collect and review information about the hazards present or likely to be present in the workplace.
- Conduct initial and periodic workplace inspections of the workplace to identify new or recurring hazards.
- Investigate injuries, illnesses, incidents, and close calls/near misses to determine the underlying hazards, their causes, and safety and health program shortcomings.
- Group similar incidents and identify trends in injuries, illnesses, and hazards reported.
- Consider hazards associated with emergency or nonroutine situations.
- Determine the severity and likelihood of incidents that could result for each hazard identified, and use this information to prioritize corrective actions.

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Computer Aided Hazards Analysis

Process simulation

- Detect unsafe temperature, pressure or flow conditions
- Identify runaway reactions

Risk assessment tools

- HAZOP, FMEA, Fault Tree Analysis
- Probability & consequence analysis

Fire & explosion modelling

- Predict blast radius
- Gas leak dispersion
- Fire spread simulation

Data & AI support

- Analyze past accident data
- Predict high-risk operating conditions
- Real-time safety monitoring

"Don't wait for accidents, simulate them."

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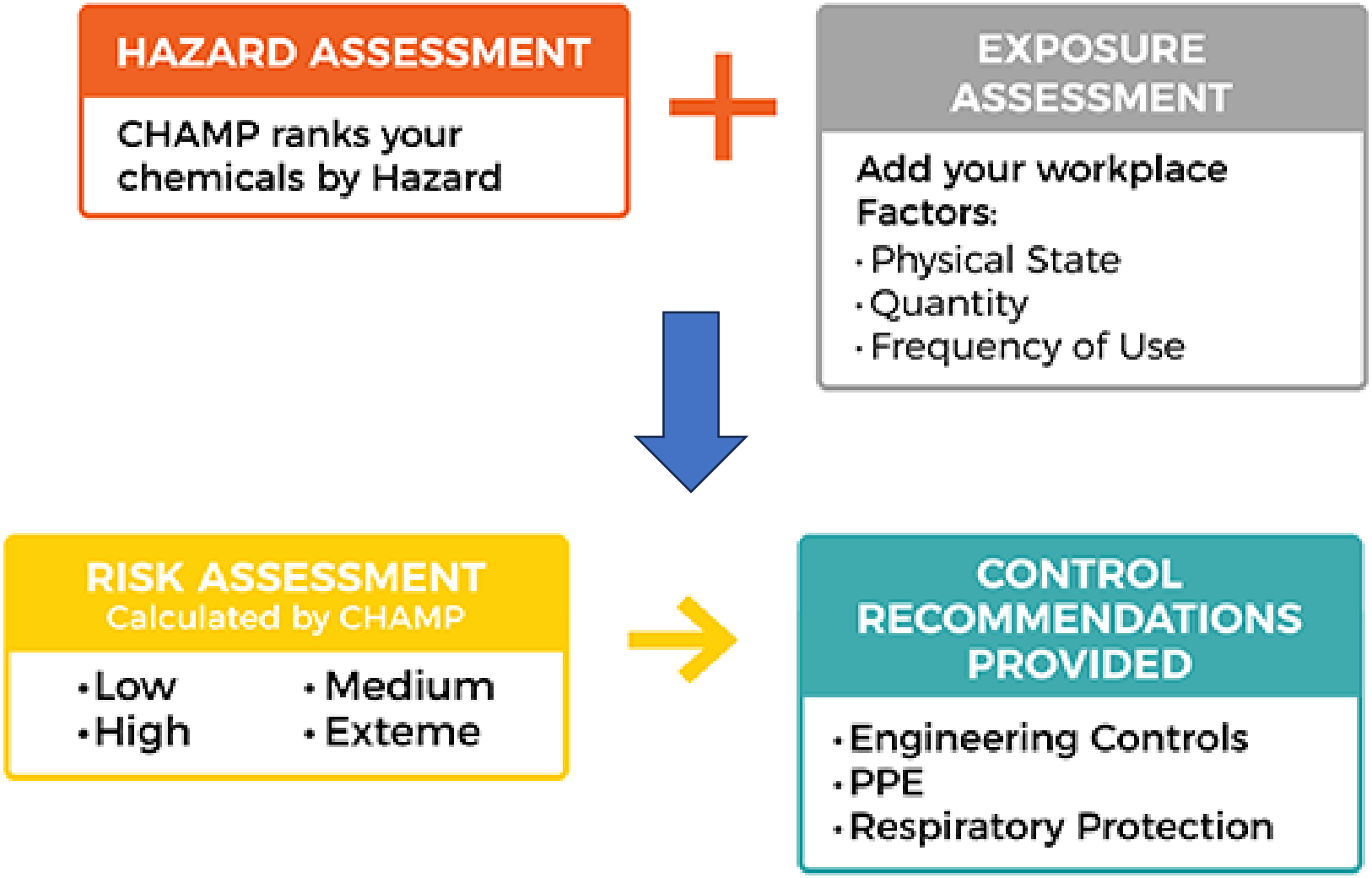
Computer Aided Hazards Analysis



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Basics of Fires and Explosion

What is Fire?

Fire is a chemical reaction that occurs when the atoms in a fuel react with oxygen. The reaction produces different molecules, such as carbon dioxide and water, and releases energy in the form of light and heat

Fuel + oxidant → oxidized fuel elements + heat + light

- Flames are primarily made up of carbon dioxide, water vapor, oxygen, and nitrogen. If the gases are hot enough, they may become ionized to produce plasma.

- The color of the flame and the fire's intensity will vary depending on the substances alight, and any impurities outside

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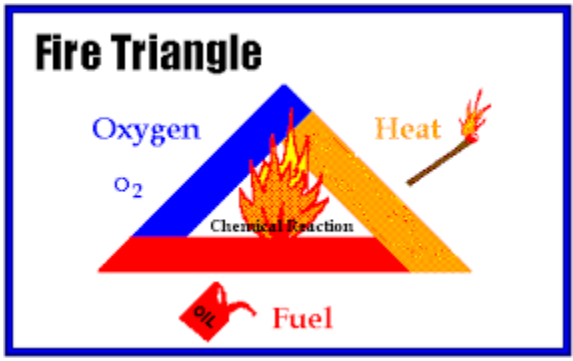
The Fire Triangle

Four things must be present at the same time in order to produce fire



- Enough oxygen to sustain combustion,
- Enough heat to raise the material to its ignition temp
- Some sort of fuel or combustible material
- The exothermic reaction

Fire will not occur if:

- Fuel is not present or is not present in sufficient quantities
- Oxidizer is not present or is not present in sufficient quantities
- The ignition source is not energetic enough to initiate the fire

Basics of Fires and Explosion

- The major distinction between fires and explosions is the rate of energy release.
- Fires release energy slowly, while explosions release energy very rapidly, typically on the order of microseconds.
- Fires can also result from explosions and explosions can result from fires.

Fire extinguishers put out fire by taking away one or more elements of the fire triangle

Types of Fires and Explosion

Class A	Ordinary solid combustibles such as paper, wood, cloth and some plastics.
Class B	Flammable liquids such as alcohol, ether, oil, gasoline and grease, which are best extinguished by smothering.
Class C	Electrical equipment, appliances and wiring in which the use or a nonconductive extinguishing agent prevents injury from electrical shock. Don't use water.
Class D	Certain flammable metallic substances such as sodium and potassium.

Fire extinguishers work by interrupting the flame's propagation. They can be used on fires involving flammable liquids, gases, and electrical equipment.

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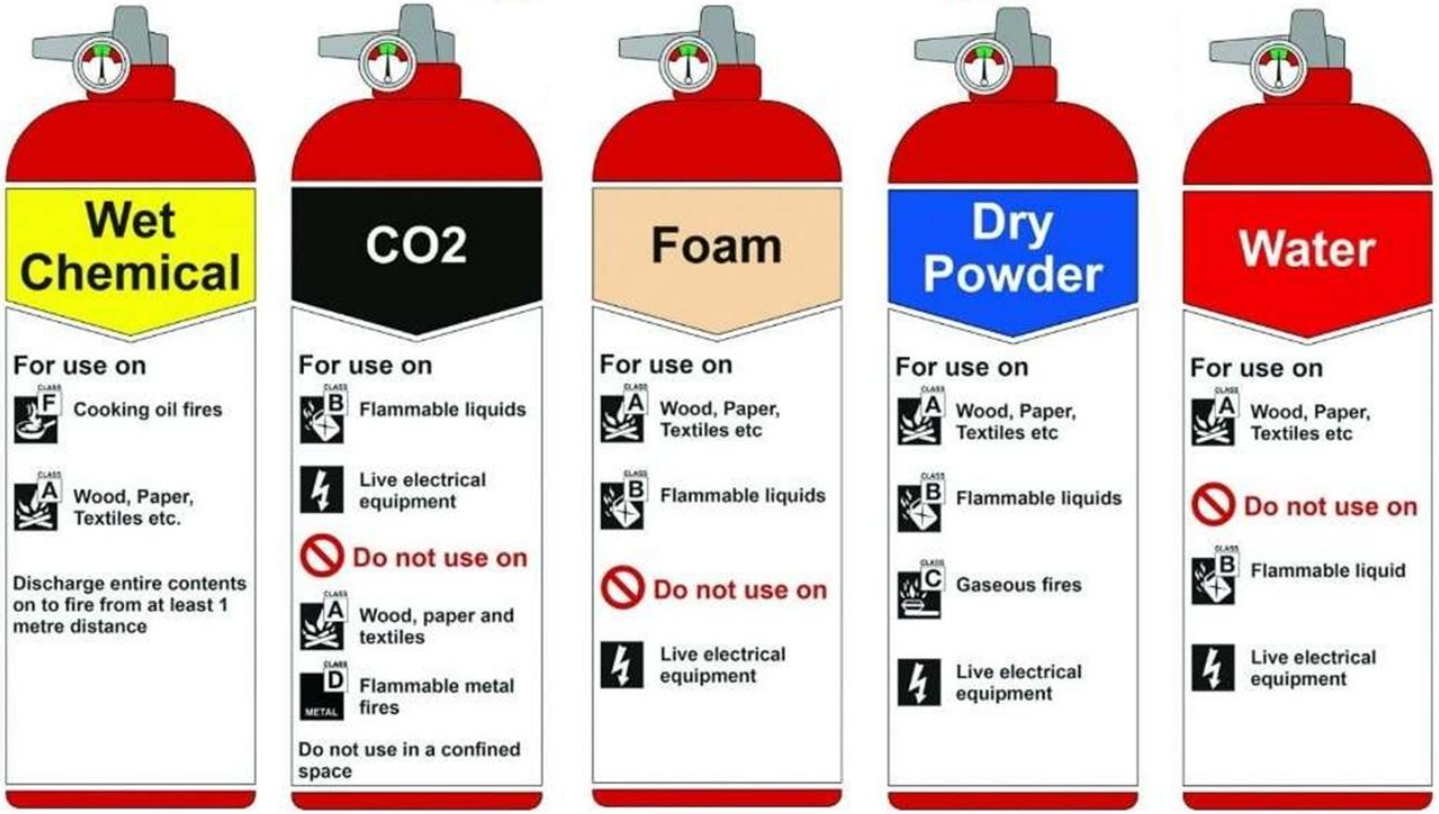
Types of Fire extinguishers

Type A	Pressurized water to be used on Class A fire only. Do not use on Class B or C fires; may cause fire spread or electrical shock
Type ABC	Dry chemical effective on all classes of fires
Type BC	Carbon dioxide to be used on chemical or electrical fires
Type K	Used in kitchens on grease fires



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Fire extinguishers

1. Water Type Fire Extinguisher

Principle: Water cools the temperature of substance that is in combustible form than its Ignition Point

Advantages

- Water fire extinguishers are cost-effective as they are cheap to purchase and to refill.
- Rapid reduction of flames due to the water extinguisher's cooling properties.
- Friendly for the environment and safe for the operator as water extinguishers contain only water and air.

Disadvantages

•Limited use as it is only recommended for class A fires (paper, wood, and textile materials).
•Cannot be used on electrical fires.
•Possible chance of freezing. As it uses plain water it is possible for it to freeze in low temperatures.

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Fire extinguishers

2. Foam Type Fire Extinguisher:

Principle: It restricts the oxygen processing to the combustible material or substance which is firing/Burning at the moment. It contains chemicals like aluminum sulphate, sodium bicarbonate and saponin.

$$Al2SO_4 + NaHCO_3 \text{—->} Al\ (OH)_3 + CO2 + Na_2SO_4$$

- The producing carbon dioxide is non-combustible. Addition of Saponin reacted with carbon dioxide to produce foam appearance which cover the Combustible material for cut off of Firing by means of restriction in supply of Oxygen to the firing Substance

Pros and Cons of foam fire extinguishers:

Pros: non-toxic, non-damaging to most materials, designed to prevent the re-ignition fires, lighter in weight than the equivalent water extinguisher.

Cons: damaging to electrical appliances. Dangerous if used on cooking fires or flammable gas fires.

Fire extinguishers

3. Dry powder extinguishers:

- Dry powder fire extinguisher smothers fires by forming a barrier between the fuel and the source of oxygen.
- It works on most types of fire; they suppress fires very quickly; specialist dry powder extinguishers are the only extinguishers that work on flammable metals.

4. CO2 extinguishers:

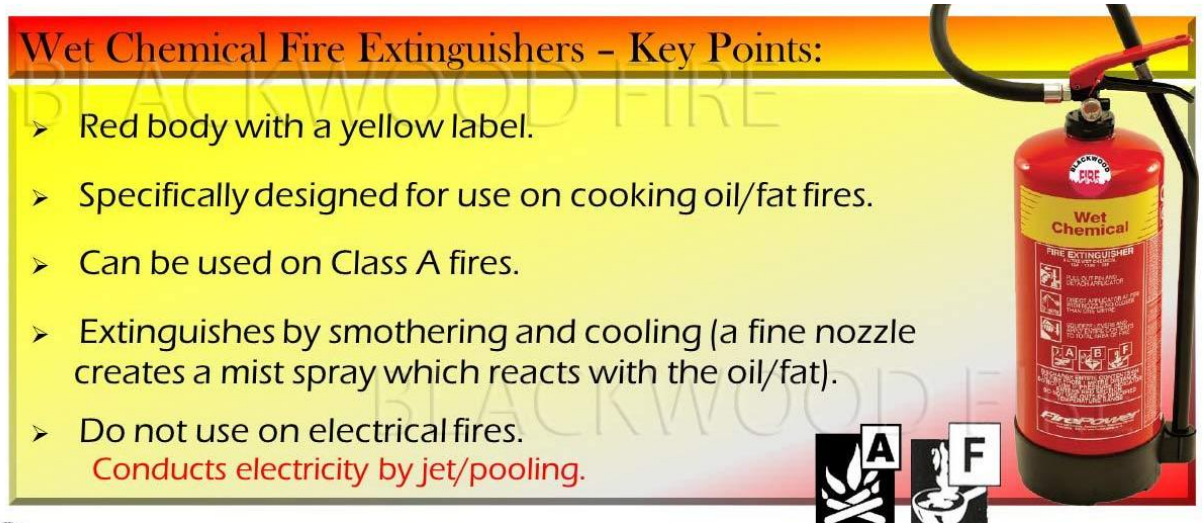
These contain only pressurised carbon dioxide gas and therefore leave no residue. Carbon Dioxide (CO2) extinguishers are used on fires involving burning liquids (Class B), and electrical fires, such as of large computer equipment, so are practical in offices. CO2 works by suffocating the fire and does not cause damage to the electrical items or cause the system to short circuit.

CO2 extinguishers are quite expensive.

Fire extinguishers

5. Wet chemical extinguishers:

Wet chemical is the only type of extinguisher to be used on kitchen (F-Class) fires. These extinguishers work by instigating a chemical reaction with fats and oils. This reaction reduces the temperature and eliminates the heat, causing flames extinguishment.



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Accident Investigations

Accident investigation is the process of determining the root causes of accidents, on-the-job injuries, property damage, and close calls in order to prevent them from occurring again.

Develop an accident investigation process that focuses on:
o fact finding, not fault finding
o determining the root causes of why the event occurred
o making changes so the event does not happen again.

Set a policy that accidents and close calls (large and small) will be investigated with equal vigor.
• Provide training and tools to staff conducting accident investigations.
• Audit completed investigations to ensure they are being completed on a timely basis with an adequate level of detail.

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Case Study 1: Union Carbide, Bhopal, India, Midnight of December 2–3, 1984

The disaster

Bhopal disaster was caused by the accidental release of 40 tones of highly toxic and heavier than air MIC gas from a Union Carbide pesticide plant located in the heart of the city of Bhopal. In addition to MIC, hydrogen cyanide and phosgene gases were believed to have been released.

MIC was used as an intermediate in the manufacture of a variety of pesticides

The scientific reason behind the accident was that water entered the tank where about 40 cubic meters of MIC was stored
When water and MIC mixed, an exothermic chemical reaction started producing a lot of heat
As a result the safety valve of the tank burst because of the increase in pressure

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Case History 1: Union Carbide, Bhopal, India, Midnight of December 2–3, 1984

Synthesis of Carbamate Pesticides

1. Petroleum coke (2C) was reacted with oxygento produce 2CO

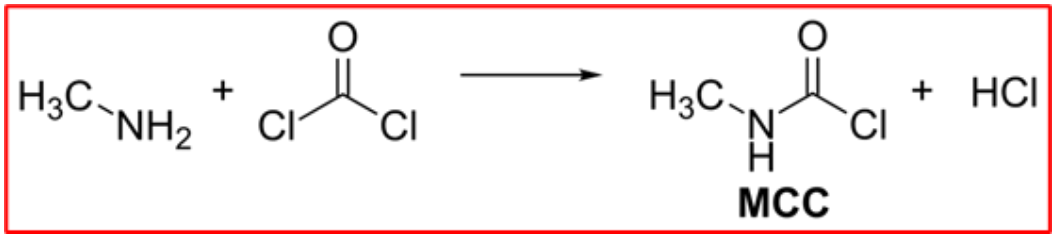
$$2C + O_2 \longrightarrow 2CO$$

2. CO and chlorine were reacted to produce phosgene ($COCl_2$)

$$CO + Cl_2 \longrightarrow COCl_2$$

3. Phosgene and methylamine (CH_3NH_2) were reacted to produce methylcarbamoyl chloride (MCC) and HCl.

$$COCl_2 + CH_3NH_2 \longrightarrow CH_3NHCOCl$$



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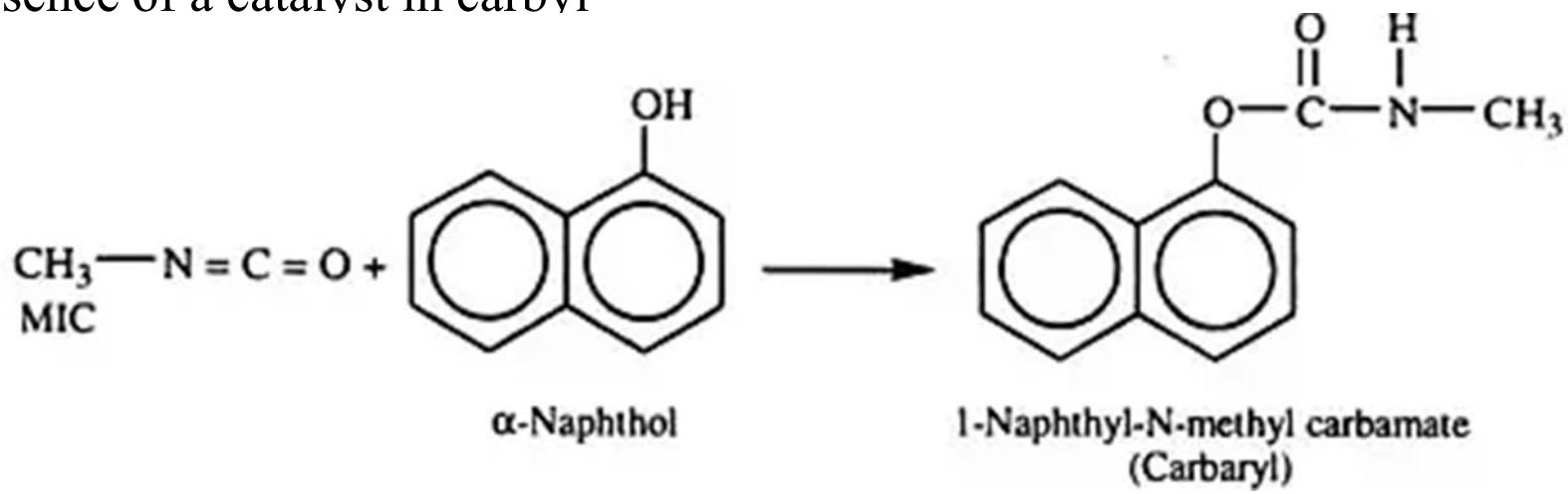
Case History 1: Union Carbide, Bhopal, India, Midnight of December 2–3, 1984

Synthesis of Carbamate Pesticides

4. Methyl carbamoyl chloride was then pyrolyzed to yield MIC (CH_3NCO) and HCl.

$$CH_3NHCOCl \longrightarrow CH_3NCO + HCl$$

5. In the last step, MIC was reacted with a slight excess of α-naphthol in the presence of a catalyst in carbyl



The Pesticide Sevin

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Case History 1: Union Carbide, Bhopal, India, Midnight of December 2–3, 1984

Careless Handling of MIC

Tanks containing MIC are required to be kept under refrigeration; this elementary caution was violated at Bhopal. The Union Carbide manual specifies that the alarm should respond whenever the temperature goes above 11°C; in Bhopal it was set at 20°C.

Failure of Safety Devices

The scrubber and flare tower, which at their best could handle minor leaks, were nonfunctional at the time of the accident.

Shoddy Maintenance There were serious lapses in the day-to-day operations

Aggravating the Tragedy

Late and misleading warning about the MIC leak increased the human cost of the disaster

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Case History 1: Union Carbide, Bhopal, India, Midnight of December 2–3, 1984

- The gas scrubber was placed on standby and therefore did not attempt to clean escaping gases with sodium hydroxide, which could have brought the concentration down to a safe level.
- The water curtain was not designed to contain a leak of such magnitude.
- The flare tower used to burn off gases before they are allowed to escape into the air was in operational, pending repairs.
- At least 16,000 people died and 150,000 to 600,000 suffered multisystemic injuries due to the exposure to the heavy gas, which rolled along ground level.
- Many of the survivors are still suffering from genetic and other disorders. The damage caused to the people is beyond quantification. Compensations were paltry ($0.5–1 billion in losses).

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Case History 2: Chernobyl Nuclear Disaster, Ukraine, April 26, 1986

- A nuclear reaction went wrong and resulted in the explosion of one of the reactors in a nuclear power plant. These reactors were constructed without containment shells.

- The release of radioactive material covered hundreds of thousands of square kilometers.

- More than 3 million people in the surrounding suburbs suffered from this disaster. While 36 people died in the accident itself, the overall death toll has been estimated at 10,000.
- The Chernobyl reactors were high-powered reactors that used graphite to help maintain the chain reaction and cooled the reactor cores with water.

- The nuclear rector operates as a "light water graphite reactor" by boiling water in fuel channels to create steam, which then powers turbines and generates electricity.

- Graphite in the nuclear reactor acts as a moderator used to help sustain the ongoing nuclear chain reaction in combination with water acting as a coolant.

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Case History 3: Fire Accident at IISER Pune (Jul 16, 2021)

- The cause of all these incidents was probably poor quality of equipment's and a lack of safety measures for the workers and also to the nearby communities.

- These incidents may be avoided by the regular checkups, overhauling, repair and maintenance, regular inspection of moving objects, electrical appliances, switches, motors, actuators, valves, pipelines, storage tanks, reactors, boilers, and pressure gauges.
- At the same time, the proper training of workers for running the operations and dealing with emergencies, spills, leaks, fire breakouts, chemical handling, and electrical shock avoidance should not be ignored.